Recap Checkpoint 3 — after 1.4 (Data Types, Data Structures & Boolean Algebra) — cumulative 1.1–1.4 — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Total = 42 marks. Award marks for valid alternatives in line with OCR positive-marking. All calculations checked below.

Q1 — [2] (AO1) — *1.1.2* - **CISC** has a **large/complex instruction set** with variable-length, multi-cycle instructions (complexity in hardware) (1). - **RISC** has a **small/simple instruction set** of fixed-length, single-cycle instructions (complexity in software/compiler) (1).

Q2 — [3] — *1.2.1* - Virtual memory uses **secondary storage as if it were RAM** (1)… - …so programs **larger than physical RAM, or more programs than fit in RAM, can run** (1). *(AO1)* - Drawback (1): **disk thrashing** — heavy paging in/out slows the system because secondary storage is much slower than RAM. *(AO2)*

Q3 — [3] (AO1) — *1.2.2* - Correct order (2; 1 mark if one pair transposed): **lexical analysis → syntax analysis → code generation → optimisation**. - The stage that removes redundant instructions / improves efficiency is **optimisation** (1).

Q4 — [3] (AO2) — *1.3.1* - Key difference (1): **hashing is one-way / irreversible**; **encryption is two-way / reversible** with a key. - Passwords are **hashed** because they **never need to be recovered** — login only compares hashes, so a stolen database does not reveal plaintext (1). - Transmitted data is **encrypted** because the **recipient must recover (decrypt) the original data**, which requires a reversible process (1).

Q5 — [3] — *1.4.1* (a) 89 = 64 + 16 + 8 + 1 → **`0101 1001`** (1). *(AO2)* (b) Nibbles `0101` = 5, `1001` = 9 → **0x59** (1 for nibble split, 1 for answer = 2). *(AO2) Check: 5 × 16 + 9 = 80 + 9 = 89 ✓.*

Q6 — [4] (AO2) — *1.4.1* (a) 39 = `0010 0111`; 52 = `0011 0100` (1). (b) −52: invert `0011 0100` → `1100 1011`, add 1 → **`1100 1100`** (1). (c) Add `0010 0111 + 1100 1100`:

```
  0 0 1 0 0 1 1 1   (39)
+ 1 1 0 0 1 1 0 0   (-52)
-----------------
  1 1 1 1 0 0 1 1
```

Result `1111 0011` has MSB = 1 → negative; magnitude = invert + 1 = `0000 1100 + 1` = `0000 1101` = 13, so the answer is **−13** (1 for the addition, 1 for converting back to denary). *Check: 39 − 52 = −13 ✓.*

Q7 — [3] — *1.4.1* (a) Mantissa must start `0.1…`; `0.0001101` so shift the point **right by 3** places → `0.1101` (1). Moving the point right by 3 **reduces the exponent by 3**: 4 − 3 = **1**, giving **`0.1101 × 2¹`** (1). *(AO2)* (b) Normalising removes wasted leading zeros so the mantissa holds the **maximum significant**

figures / greatest precision for the available bits (1). *(AO1) Examiner tip: moving the point right DECREASES the exponent; the value is unchanged.*

Q8 — [3] (AO2) — *1.4.3* - Distribute (distributive law): A·(Ā + B) = (A·Ā) + (A·B) (1). - A·Ā = **0** (complement / annihilation law) (1). - 0 + (A·B) = **A·B** (identity law) (1). Final answer: **A·B**.

Q9 — [4] (AO2) — *1.4.3*

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

(Working: `A+B` is 0,1,1,1; `¬(A·B)` is 1,1,1,0; AND them → 0,1,1,0.) *(Up to 3 marks for the F column: 1 per correct pair of rows, plus 1 if the whole column is correct; allow follow-through.)* F is 1 only when the inputs **differ**, so it is equivalent to an **XOR** gate (1 mark for identifying XOR).

Q10 — [4] (AO3) — *1.4.2* (a) First `pop` returns **5**; second `pop` returns **9** (1 + 1). Trace: push 8 → [8]; push 3 → [8,3]; push 5 → [8,3,5]; **pop** → **5** → [8,3]; push 9 → [8,3,9]; **pop** → **9** → [8,3]. (b) Stack contents bottom → top: **8, 3** (1). A stack is **LIFO (Last In, First Out)** — the **most recently pushed item is the first popped**, as both pops returned the most recent pushes (1).

Q11 — [4] (AO3) — *1.4.2* (a) BST from inserting 60, 40, 80, 30, 50, 70:

```
        60
       /  \
     40    80
    /  \   /
  30   50 70
```

(1 for 60 root with 40 left / 80 right; 1 for 30 & 50 under 40 and 70 as left child of 80.) (b) In-order (left, root, right): **30, 40, 50, 60, 70, 80** (1). This is **ascending sorted order**, demonstrating the BST ordering property (left subtree < node < right subtree) (1).

Q12 — [6] (AO3) — *synoptic 1.1 + 1.4 + 1.3* Mark holistically; up to 6 from a reasoned evaluation covering **all three** decisions: - **Processor:** recommend **RISC** (1) — fixed-length, single-cycle instructions give **low power consumption and predictable, fast real-time response** suited to a battery embedded device; *(reject CISC unless its fewer-instructions benefit is justified against the low-power need)* (1 for justification). - **Data structure:** recommend a **(circular) queue** (1) to buffer the **most recent readings in time order (FIFO)** with fixed memory and O(1) enqueue/dequeue; *(accept a justified stack/array only with sound reasoning)* (1 for justification). - **Compression:** recommend **lossless** (1) because **medical readings must not lose accuracy/data**, so lossy (which discards data) is unsuitable despite smaller files (1 for justification linking to clinical accuracy). *Indicative full-mark answer weaves RISC (power/real-time) + circular queue (recent readings, FIFO, fixed memory) + lossless (data integrity), each justified against the heart-rate monitor's requirements.*